

Flash games on the Web or as game apps within Facebook. This wouldn't be possible with HTML5 anytime soon.

Mobile applications

The *smart* phone

Today, there are more than 5 billion people around the world using mobile phones. No other technology is advancing at the rapid speed the mobile industry is experiencing. A new category of mobile phone is rapidly growing as well, the smart phone. A few years ago, a smart phone was something that maybe just allowed you to send an email. Today, when you think smart phone, you think email, web, games, MMS, video conferencing, basically a computer in your pocket. There are a number of companies leading the next wave of smart phone market. Google, Apple, RIM, Nokia, Microsoft, and HP all have their own operating systems and hardware. Consider this, at the end of 2009, a mobile phone running at 500 MHz with a 3 MB camera was considered screaming fast. Now, if you want something good you go for one with 1 GHz with a 1 GB of RAM, an 8 MP camera, front and rear facing cameras, proximity devices and sophisticated operating systems (OS) that rival, and in some cases exceed, what you can accomplish on your desktop.

Building Applications for Mobile Devices

With a global figure of 5 billion mobile users, it is clear that the smart phone market has massive potential for growth. So, what does it mean to develop for a smart phone? At the end of the day, there are essentially two ways you can develop for a smart phone:

- ▶ Develop directly to the software development kit (SDK)
- ▶ Develop using an intermediate technology

Each mobile device these days comes with an SDK that you can use for development. An SDK comes with the development tools, bundling tools, and emulators you need to test your code. When you need access to the latest and greatest technology, you need to use an SDK. The challenge you have with using core SDKs is that you need to use the native development language. This is different for each SDK. For instance, Apple prefers you use Objective-C whereas Google prefers you use Java.

The second way to develop mobile devices is to use an intermediate technology that allows you to build for multiple devices using only one language. An example of this is the 3D game development technology called Unity 3D. Unity uses JavaScript to let you to script your games